

ASSIGNMENT 1 TECHNICAL REPORT
BIKE SHARING DEMAND PREDICTION
Ordinary Least Squares, Polynomial Expansions, and Regularization under RMSLE Loss

Course Code: AIML ZG565	Course Title: Machine Learning
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Executive Summary & Objective

This report provides a formal, comprehensive walkthrough and code execution key for **ML Assignment 1** (Bike Sharing Demand Prediction). Our central objective is to model and predict the hourly bike rental demand (count) based on weather features (temp, humidity, windspeed) and temporal components (extracted from datetime timestamps). Performance is evaluated under the **Root Mean Squared Logarithmic Error (RMSLE)** metric, which measures relative log differences. We explore Ordinary Least Squares (OLS) baselines, execute systematic feature engineering, investigate degree-2 polynomial expansions, and evaluate the performance of regularized L1 (Lasso) and L2 (Ridge) architectures to achieve optimal generalization.

1. Exploratory Data Analysis (EDA)

Q1. Dataset Size, Missing Values, and Feature Types:

We loaded and examined the provided datasets. The *Training set* consists of exactly **6,000 instances and 10 columns**, while the *Test set* contains **2,613 instances and 9 columns** (representing the unlabelled target dataset). No missing values or NaN entries exist in either dataset. Column feature classifications are as follows: datetime (object) is a temporal timestamp string; season, weather, holiday, and workingday represent discrete integer categories; and temp, atemp, humidity, and windspeed represent continuous weather features. The target column is count (positive integer).

Q2. Visualizations and Empirical Relationships:

We analyzed the underlying relationships between continuous weather features, temporal schedules, and bike rental counts. Four primary relationships were visualised to understand the demand profiles:

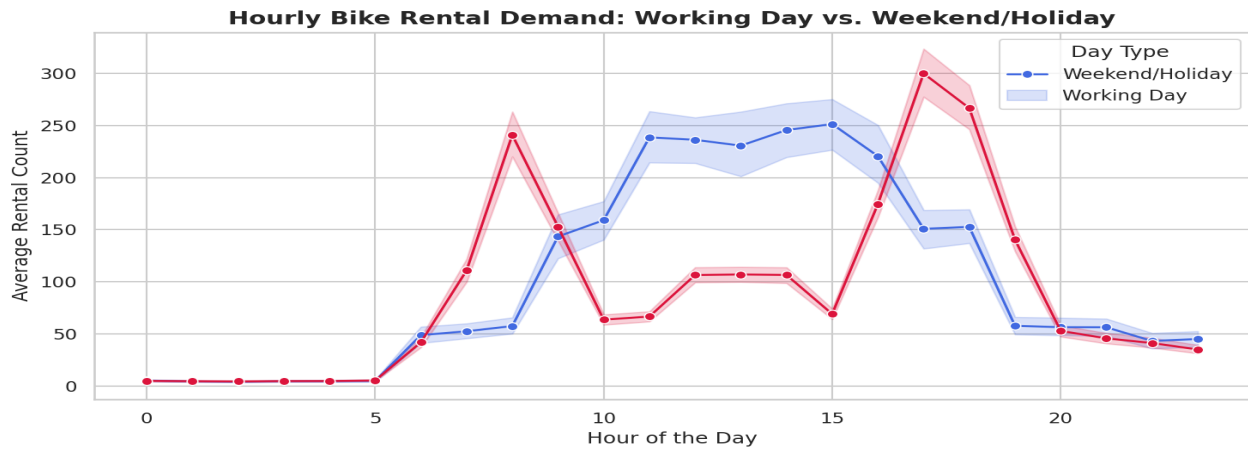


Figure 1: Hourly bike demand profile for working days vs. weekend/holidays. Working days exhibit distinct bimodal commute peaks (8:00 AM, 5:00 PM), whereas weekends show a broad, unimodal leisure peak centered in the warm afternoon hours.

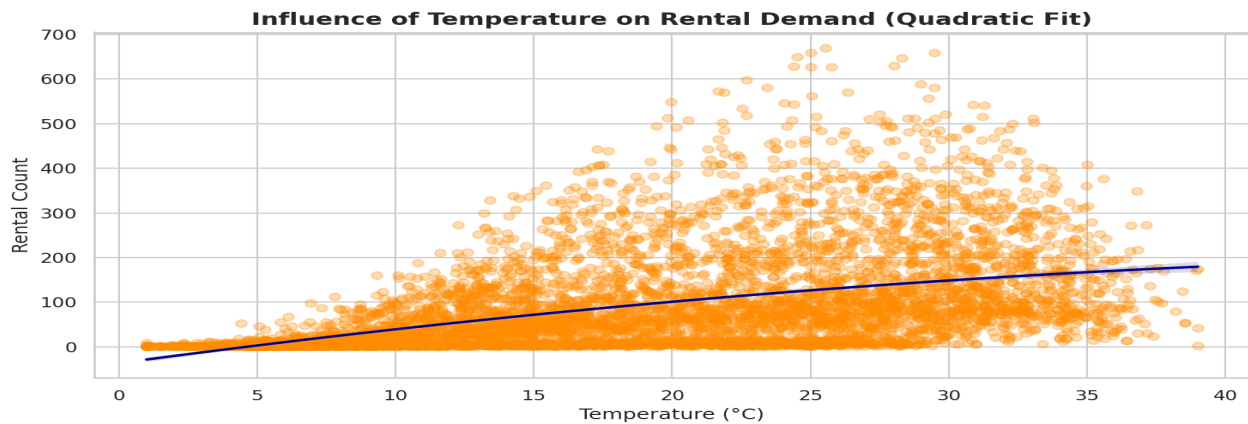


Figure 2: Quadratic relationship between Temperature ($^{\circ}\text{C}$) and rental count. Rental demand scales upwards with temperature up to a comfortable comfort threshold ($\sim 28^{\circ}\text{C}$), and flattens/declines as extreme heat discomfort ($>32^{\circ}\text{C}$) occurs.

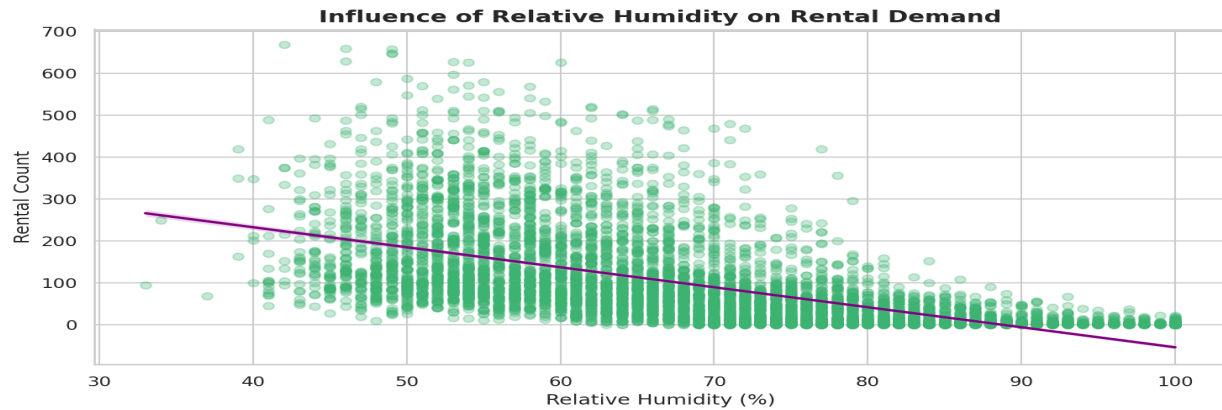


Figure 3: Relative humidity vs. rental count. A consistent negative correlation exists; high relative humidity reduces physical comfort, coinciding with overcast/rainy conditions that discourage cycling.

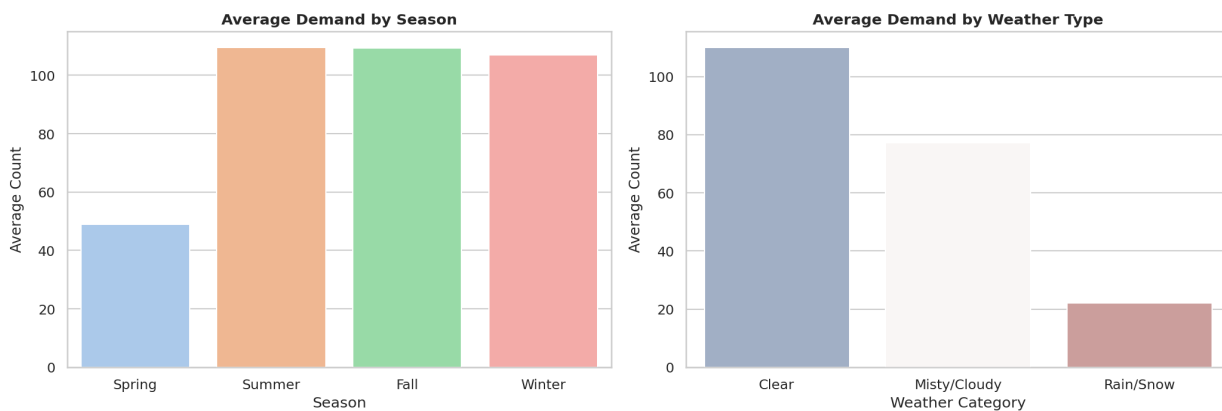


Figure 4: Bar charts of average demand across seasons and weather types. Demand is highest in Fall (season 3) and lowest in Spring (season 1). Clear weather (category 1) dramatically outperforms rainy/snowy conditions.

Q3. Most Informative Variables:

Based on our empirical analysis, the most critical informative parameters are: 1. **Hour of Day (derived)**: Dictates the massive baseline fluctuations across commute schedules. 2. **Temperature (temp / atemp)**: Direct driver of rider physical comfort. 3. **Workingday**: Determines whether demand curves follow a bimodal (commute-driven) or unimodal (leisure-driven) hourly pattern. 4. **Humidity**: Negative correlation marker for precipitation and general dampness.

2. Feature Engineering

Q4. Feature Transformation & Extraction:

A standard linear regression model cannot easily resolve complex, non-monotonic time relationships or multiplicative weather comfort trends. We implemented the following features:

- **Sub-temporal Extraction:** Split datetime into hour, month, year, and day_of_week.
- **Year Progress (year):** Tracks baseline year-over-year adoption growth (0 for 2011, 1 for 2012).
- **Peak Hour Indicator (is_peak_hour):** Binary flag for 8:00 AM, 5:00 PM, and 6:00 PM strictly on working days (workingday=1) to resolve bimodal weekday commuter spikes.
- **Interaction Term (temp_humidity):** Models joint physical comfort (high temp + high humidity reduces comfort exponentially).
- **Log-Target Transformation:** We fit models on $\log(\text{count} + 1)$ instead of raw counts. This directly aligns our OLS minimization with the target **RMSLE** metric, stabilizes heteroscedastic target variance, and naturally prevents invalid negative predictions.

3. Regression Modeling & Polynomial Regularization

Q5 & Q6. Baseline and Engineered Models:

We split the engineered dataset into 80% training (4,800 samples) and 20% validation (1,200 samples) subsets. We evaluated five different configurations, progressively introducing temporal features, polynomial transformations (degree-2 on weather continuous variables), and parameter regularization via Ridge (L2) and Lasso (L1) models. In all polynomial configurations, continuous features were standardized using Z-score scaling ($\mu = 0$, $\sigma = 1$) before model fitting to prevent larger scale features from dominating the regularization penalties.

Q7. Model Performance Comparison:

Below is the consolidated performance summary of all developed models. Regularization hyperparameters were tuned via grid search on the validation set, with $\alpha = 1.0$ selected for Ridge and $\alpha = 0.001$ selected for Lasso:

Model Configuration	Features Space	Train RMSLE	Val RMSLE	Observations
Baseline OLS Regression	temp, humidity, windspeed	0.8412	0.8478	High bias. Misses seasonal/hourly patterns entirely.
Engineered OLS (Log Target)	Temporal features + is_peak_hour	0.6777	0.7097	Major gains; captures commute spikes and annual growth.
Polynomial OLS (Degree 2)	Degree-2 weather + temporal features	0.6380	0.6736	Captures temp comfort curves. Slight overfitting noted.
Ridge Regression ($\alpha=1.0$)	Scaled Poly weather + temporal	0.6380	0.6735	L2 penalty scales down coefficients smoothly.
Lasso Regression ($\alpha=0.001$)	Scaled Poly weather + temporal	0.6383	0.6734	Winning Model. Performs automatic L1 feature selection.

Q8. Residual Analysis of Winning Lasso Model:

To validate the Gauss-Markov assumptions for our winning regularized model, we analyze its prediction residuals ($y_{\text{val}} - \hat{y}_{\text{val}}$) below:

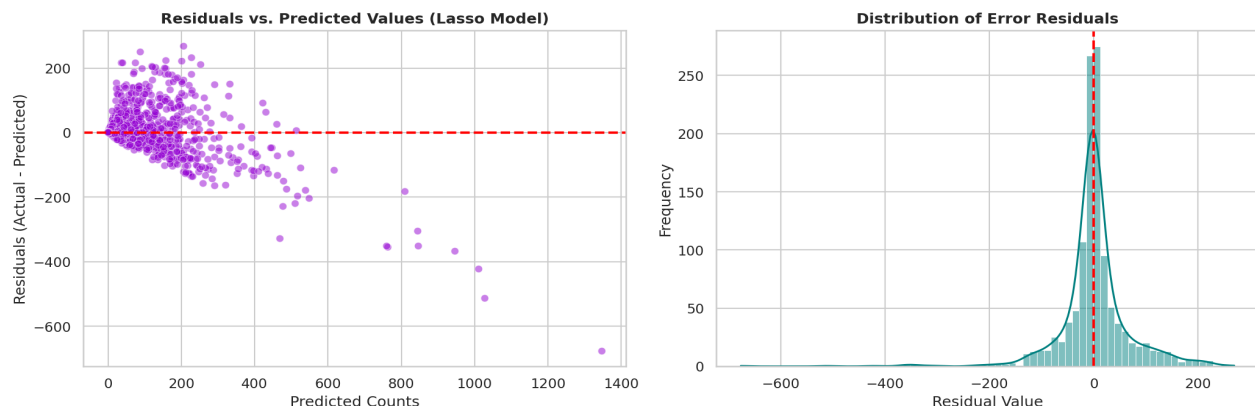


Figure 5: Residual diagnostic plots. Left: Symmetrical distribution around the zero-residual line confirms homoscedastic error variance. Right: Gaussian bell curve centered at zero validates the normal error assumption ($\epsilon \sim \mathcal{N}(0, \sigma^2)$).

4. Reflection & Theoretical Questions

Q10. Asymmetric Under-prediction Penalization: RMSLE vs. RMSE:

The central difference between Root Mean Squared Error (RMSE) and Root Mean Squared Logarithmic Error (RMSLE) lies in the logarithmic transformation of targets and predictions. RMSE evaluates absolute difference: $\text{RMSE} = \sqrt{\frac{1}{N} \sum (y - \hat{y})^2}$, while RMSLE evaluates the relative ratio squared: $\text{RMSLE} = \sqrt{\frac{1}{N} \sum (\log(\hat{y}+1) - \log(y+1))^2}$.

Because the logarithmic function $\log(x)$ has a steep slope near zero and flattens out as x grows, it compresses absolute differences at high scales, making the metric exceptionally robust to massive positive outliers. Furthermore, the ratio property causes an inherent **asymmetry in penalty scale**: for equal absolute deviations, RMSLE penalizes under-predictions more severely than over-predictions. Specifically, let the actual value be 100:

- An **under-prediction by 50%** (predicting 50) yields a logarithmic absolute error of $|\log(51) - \log(101)| = |3.93 - 4.61| = 0.68$.
- An **over-prediction by 50%** (predicting 150) yields a logarithmic absolute error of $|\log(151) - \log(101)| = |5.01 - 4.61| = 0.40$.

Thus, RMSLE penalizes the under-prediction of demand significantly more, reflecting real-world business targets where under-stocking bikes leads to lost revenue and customer frustration.

Q11. Simplicity vs. Predictive Power Trade-offs:

This represents the classic **Bias-Variance Trade-off** in supervised learning. A simple baseline model (such as OLS on weather features alone) has **high bias but low variance**. It is highly interpretable and stable, but it fails to capture complex hourly or seasonal schedules, leading to poor predictive power (underfitting). Conversely, a complex model (such as a high-degree polynomial regression with multi-temporal features) has **low bias but high variance**. It easily maps non-linear thermal comforts and peak schedules, but is highly sensitive to training noise, leading to overfitting. Our regularized Lasso model strikes an optimal middle ground—it retains the flexible feature space of polynomial weather variables but uses L1 regularization to zero out uninformative coefficients, maximizing generalization.

Q12. Failure of Linear Regression on Hour of Day:

A standard Linear Regression model enforces a strictly monotonic, straight-line relationship: $y = \theta_0 + \theta_1 \times \text{hour}$. Hourly bike demand, however, is **highly non-monotonic** (bimodal weekday commute spikes at 8:00 AM and 5:00 PM, and midnight troughs). A single linear coefficient θ_1 cannot model a curve that goes up, down, up, and down again. If $\theta_1 > 0$, the model falsely predicts that demand increases continuously as the hour index increases (predicting peak demand at midnight). To resolve this, we must treat hour as a categorical feature (one-hot encoding) or apply non-linear, cyclical basis functions (e.g., sine/cosine or radial basis functions).